Codebusters: Practice Test 1

Time: 50 minutes · 5330 points. Bring only pencils and erasers; the Event Lead gives each pair a resource sheet with the Baconian table and the Galactic sentence (the Vigenere letter numbers are printed in Part 4). The PigPen, Tap Code, and Knight's Templar keys are NOT given, so draw them from memory first. Split the questions with your partner and do the fastest ones first: 2 or fewer mistakes on a question still earns full points, each extra mistake loses 50, and most teams do not finish every question.

Partner names: _____ and _____ Score: _____

Part 1: Caesar (shift) ciphers

Every letter moved the same number of places, 1, 2, or 3, forward or back. Find the shift, then decode.

1. The shift was +2 (each letter moved 2 places forward). Decode: JQPGADGGU VCNM DA FCPEKPI (100 pts)

2. Find the shift, then decode: RKNSGR ZQD FNNC RVHLLDQR (125 pts)

3. Find the shift, then decode: VKDUNV KDYH QR ERQHV (125 pts)

4. Find the shift, then decode: QEB JLLK EXP KL TFKA LO OXFK (150 pts)

5. Find the shift, then decode: JGEFRLGLE GQ FMRRCP RFYL RFC QSPDYAC MD RFC QSL (175 pts)

Part 2: Atbash ciphers

The alphabet is flipped: A=Z, B=Y, C=X, and so on. Write A to M over Z to N to make your key.

6. Decode: RXV UOLZGH LM DZGVI (100 pts)

7. Decode: YZGH FHV VXSLVH GL URMW ULLW (125 pts)

8. Decode: HVZ LGGVIH FHV ILXPH ZH GLLOH (125 pts)

9. Decode: NZIH RH XZOOVW GSV IVW KOZMVG (150 pts)

Part 3: Aristocrats

Each letter was swapped for a different letter, and no letter stands for itself. Spaces are kept. Count letters, look for short words like A, THE, and AND, and use the hint when there is one.

10. Hint: the first word is EARTH. Decode: ZQVRP IDKAI CASZ ZMZVU HQU QAH RVQMZWI QVCOAH RPZ IOA CASZ ZMZVU UZQV (200 pts)

11. Hint: the last word is CHRYSALIS. Decode: Y SYRILTMEEYL RGLFD MFRP Y ZGRRILKEV MFDMHI Y SULVDYEMD (250 pts)

12. Hint: the word MAMMALS is in the message. Decode: LCZWGY ZOB BQWNCPOY ZMG AZAAZWY YQ XCGV ARYX HQAG RN XQ UMGZXCG ZPM (250 pts)

13. Hint: the word BLOOD is in the message. Decode: ULV LVGYU ON G ZENSKV ULGU DEZDN RKHHC GKK GYHEFC ULV RHCQ (300 pts)

14. No hint. Decode: QBTYCV HVN VHYBWEXC TYP RTCNG TYP TWG CJ LTDN CXNWG JRY IJJP TYP CXNM EWZN JII CXN JKMENY CXTC RN SGNTCXN (350 pts)

15. No hint. Decode: LWPT GTUSWWU AGTTL EGT AEIITG AQER E AQOGAF LAWGF JCOIUORN ERU AQTF ZER IOHT DWG ASW AQWCLERU FTEGL (400 pts)

Part 4: Vigenere ciphers

Letter numbers: A=0 B=1 C=2 D=3 E=4 F=5 G=6 H=7 I=8 J=9 K=10 L=11 M=12 N=13 O=14 P=15 Q=16 R=17 S=18 T=19 U=20 V=21 W=22 X=23 Y=24 Z=25. Write the key under the message again and again (skip spaces). Cipher number minus key number; if it is below 0, add 26.

16. The key is SKY. Decode: UVMMNQ SBC EKBW YD LSLQ NPGZQ (175 pts)

17. The key is BUG. Decode: BHZT WGO FOGN NFUBZ FUBXY (175 pts)

18. The key is MOON. Decode: AQSNZ HWQQG FVES OAP TOYX HKVOS O QMM (225 pts)

Part 5: Tap Code

Each number is row, then column, in the 5 by 5 Tap Code grid (43 means row 4, column 3). / means a new word. The grid is not given, so draw it first. C and K share a box.

19. Decode: 43 15 15 14 43 / 43 35 42 34 45 44 (125 pts)

20. Decode: 13 15 15 35 / 44 42 54 24 33 22 (150 pts)

21. Decode: 13 34 42 11 31 / 24 43 / 11 33 / 11 33 24 32 11 31 (175 pts)

Part 6: Baconian ciphers

Each group of 5 symbols is one letter. Sort the symbols into an A group and a B group, write the A's and B's, then use the Baconian table on the resource sheet (I and J share, U and V share).

22. Hint: + and = stand for A. # and % stand for B. Decode: +#=#% =%%+# =#%=# =%#+= (150 pts)

23. Hint: the first letter is W. Decode: ><>(< ()<<< ())((<<()) (200 pts)

24. No hint. Decode: &@@@& @$$@@ @$$@& &@&@@ (225 pts)

Part 7: PigPen

The PigPen key is not given, so draw it from memory first. In the first two PigPen questions, each symbol is described in words: which sides have lines, or which way the V shape points, and whether it has a dot.

25. Decode this 4-symbol word: (1) V shape (point down). Dot. | (2) Lines on right, bottom. No dot. | (3) ^ shape (point up). No dot. | (4) Lines on all 4 sides. No dot. (125 pts)

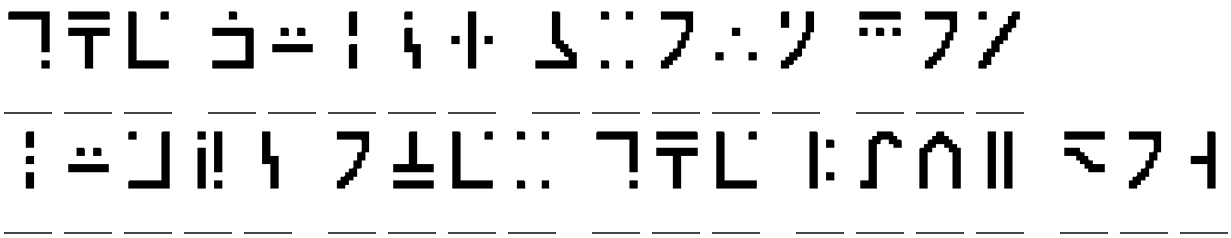
26. Decode this 5-symbol word: (1) > shape (point right). No dot. | (2) Lines on top, left. No dot. | (3) Lines on top, right. No dot. | (4) Lines on all 4 sides. No dot. | (5) Lines on top, left. Dot. (150 pts)

27. Draw the PigPen symbols for the word STORM. (150 pts)

28. Draw the whole PigPen key from memory (two grids and two X's). (100 pts)

Part 8: Knight's Templar and Galactic Alphabet

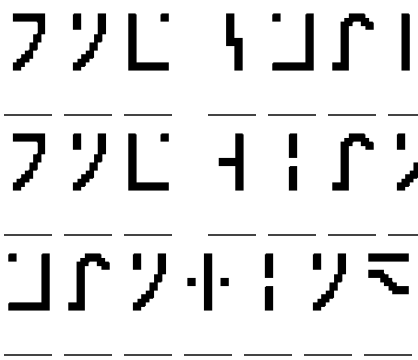
The Knight's Templar key is not given at the tournament. The Galactic Alphabet is given on the resource sheet only as this sentence, THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG (Y is always ||). Label it first:



29. Draw the whole Knight's Templar key (A to Z) from memory. (100 pts)

30. To find the Galactic symbols for J, V, and Z on the resource sheet, which word of THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG do you look in for each letter, and which letter of that word is it? (100 pts)

31. Solve this quote from Neil Armstrong which has been encoded with the Standard Galactic Alphabet. (80 pts)



22	MOON (ABABB ABBAB ABBAB ABBAA)
23	WIND (< and (are A; > and) are B: BABAA ABAAA ABBAA AAABB)
24	SNOW (@ is A; & and $ are B: BAAAB ABBAA ABBAB BABAA)
Part 7: PigPen	
25	WAVE
26	TIGER
27	S: V shape (point down), no dot. T: > shape (point right), no dot. O: lines on top, bottom, left, with a dot. R: lines on top, left, with a dot. M: lines on top, right, bottom, with a dot. (2 or fewer wrong symbols is full credit.)
28	Grid 1: A B C / D E F / G H I. Grid 2 (dots): J K L / M N O / P Q R. X 1: S top, T left, U right, V bottom. X 2 (dots): W top, X left, Y right, Z bottom. Check it against the PigPen key card.
Part 8: Knight's Templar and Galactic Alphabet	
29	Check it symbol by symbol against the Knight's Templar key sheet from the cards file, including the dots. 2 or fewer wrong symbols is full credit.
30	J: JUMPS, 1st letter. V: OVER, 2nd letter. Z: LAZY, 3rd letter.
31	ONE SMALL STEP FOR MAN, ONE GIANT LEAP FOR MANKIND (Neil Armstrong). 2 or fewer wrong letters is full credit.